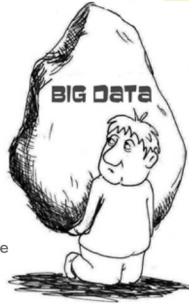


IMPACT OF BIG DATA ON ENTERPRISE INTEGRATION

This article explains the new challenges in enterprise integration caused by the advent of big data, and presents some approaches to overcome them.



Big data is the latest buzz-word in the industry. At the basic level, big data is just that: large amounts of data that cannot be handled by conventional systems. With increases in hardware capacity, our definition of what constitutes a conventional system changes, and so does the threshold of big data, which is not something new but has always been around. It's just that the threshold of what constitutes big data has changed. Today, the threshold for big data may be terabytes (10^{12} bytes). Soon, it will be petabytes (10^{15} bytes). Twenty years ago, there were very few systems that could process gigabytes (10^9 bytes) of data in an acceptable time-frame. So gigabytes would have been the lower threshold of big data at that time.

Integration

Integration is needed when we have to connect two or more software systems together. Large enterprises will have tens or hundreds of systems to connect together. Since the number of systems involved in integration could be large, the amount of data that flows across these connections will also be large.

How does big data impact enterprise integration?

With the advent of big data processors, more and more organisations recognise the value of big data analytics, and the need to process big data within enterprises is increasing. This will lead to large amounts of data (which is a subset of the big data captured by the organisation) moving across integration middleware (a software that connects two or more software systems). Such large amounts of data would overload existing integration middleware systems, since they were designed to handle lower volumes of data. This is depicted in Figure 1.

Let us analyse Figure 1 in detail. Big data is not what is really interesting. Who needs a mountain of data anyway? What we want are the results of processing this data. Depending on the processing algorithm used to process the data, we get

different results. This is where it gets interesting. Let us suppose that an enterprise implements a big data solution. The big data processing (refer to the big data processor in the figure) solution could be a shared service across the enterprise. Due to the technical complexity and cost involved in building a big data processor, it is not possible for each division in the organisation to have its own. But due to the business value accruing from big data analysis, sooner or later, different divisions in the enterprise will want their own processing on the big data set. This can be done by moving a subset of the big data, relevant to that division, over to its systems. This is the point at which the integration systems in the enterprise will feel the impact. To enable different divisions in the organisation to 'do their own stuff' with data, a subset of the big data will start moving across the integration middleware.

Overcoming the big data challenge

So how can we solve the new challenge caused by subsets of big data moving across the integration middleware? Let us look at three options:

1. Buy more hardware: This is technically called vertical scaling or horizontal scaling. This approach may work with smaller challenges, but not with big data, since the amount of new hardware required to support the load will make the idea financially unviable.
2. Buy specialised big data solutions: These solutions can be purchased and given to each division that needs a big data processor. There are very few big data processors in the market now, but we can expect more soon. This will be cheaper than buying more hardware, and can be considered, if the organisation has enough funds. Note that this approach works by avoiding the problem altogether: we avoid moving any data across the integration middleware. On the flip side, this approach will result in multiple copies of big data in the enterprise.
3. The third option involves extending our middleware's